Nullivance Propositional Logic: Threshold-Signed Consequence on the Unit Square with a Machine-Checked Completeness Theorem*

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Abstract

We present the propositional core of *nullivance logic* (NPL). Formulas take values in the unit square $[0,1]^2$ — independent degrees of support for truth and for falsity; negation swaps the channels, conjunction and disjunction act as $(\min, \max)$ and $(\max, \min)$, and a *harmonization* connective $\oplus$ acts as $(\min, \min)$: bilattice consensus as a first-class connective. A *manifestation threshold* $\tau \in (0, 1]$ induces four sign predicates T^+, T^-, F^+, F^-; consequence is defined uniformly in the threshold and shown threshold-invariant (a well-definedness property, not an added strength). Main results, each machine-checked in Lean 4 unless noted: (i) thresholding commutes exactly with evaluation, projecting the continuous semantics onto the four-valued matrix **4**; (ii) a four-signed analytic tableau calculus is sound and complete for signed consequence, over **4** and, via the exact projection, over the continuous semantics; consequence is decidable, compact and strongly complete for infinite premise sets; (iii) NPL is paraconsistent — explosion, disjunctive syllogism, and detachment for the material conditional all fail — while harmonized contradictions $\varphi \oplus \neg\varphi$ collapse to an unmanifest state; (iv) $\oplus$ is not definable from the $\{\neg, \wedge, \vee\}$ fragment, and the natural-deduction rule set of the source system is provably incomplete for it — its rules constrain only the truth channel of $\oplus$; (v) a generative tier beneath the semantics derives truth-objects from (intensity, structure) channel pairs through a replaceable stability function, with the logic provably independent of that choice. We position NPL against

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Belnap–Dunn logic, bilattice logics, and paraconsistent Gödel logic: the semantic ingredients are known, the unsigned fragment is an instance of Arieli–Avron's logical-bilattice consequence, and the contribution is the signed architecture (which provably falls outside the bifilter framework, and is provably not internalizable in the object language), its exact projection, and the verified metatheory.

1 Introduction

Two ideas drive nullivance logic.[1] First, the epistemic state of a proposition has two independent coordinates: how strongly it is supported as true, and how strongly as false. This is the two-channel reading familiar from Belnap–Dunn four-valued logic [2, 3, 7] and its continuous relatives; we inherit that literature rather than compete with it. Second, *manifestation is thresholded*: a proposition manifests as true (false) only when the corresponding support clears a threshold τ. Below both thresholds lies an unmanifest state; above both lies manifest contradiction. A designated connective $\oplus$ (*harmonization*) takes the consensus of both channels, and harmonizing a proposition with its own negation drives it into the unmanifest state instead of exploding.

Honest positioning. None of our semantic ingredients is new in isolation, and we say so precisely (Section 9): the $\{\neg, \wedge, \vee\}$ fragment of the continuous semantics coincides with the propositional base of paraconsistent Gödel logic [5] and with product-bilattice semantics [9, 8]; $\oplus$ is the bilattice consensus $\otimes$, which at the four-valued level carries complete calculi [1]; the projection core is the cutworthiness of α-cuts [12]; signed tableaux for finitely-valued logics are classical [10, 15]. Moreover — a comparison we performed rather than conjectured — the *unsigned* fragment of our consequence relation is an instance of Arieli–Avron's logical-bilattice consequence (Proposition 9.1). The contributions are:

1. **Signed consequence beyond designated values** (Definition 3.2, Proposition 9.1): four sign predicates read the two channels against the threshold. The negative signs T^-, F^- are complements, and we prove their satisfaction sets are not bifilters; the four-signed relation is therefore not a logical-bilattice matrix consequence in the sense of

[1] The name derives from the system's founding concern with *structured absence*: states that are null in intensity yet carry direction (Latin *nullus* + valence). The source material calls the pre-logical version of this notion *quasivance*; both terms are defined where they are used (Section 8) and are otherwise dispensable labels for the formal objects.

[1], while its unsigned fragment provably is. The relation is threshold-invariant (Proposition 3.3), which we state as a well-definedness result.

2. **Exact projection** (Theorem 4.2): thresholding commutes with evaluation, collapsing continuous consequence onto the finite matrix **4** without loss.

3. **A machine-checked completeness theorem** (Theorem 6.2), plus decidability (Theorem 6.3) and compactness and strong completeness (Theorem 6.5). To our knowledge no verified completeness theorem exists for a consensus-bearing logic over $[0,1]^2$ with signed consequence (search scope in Section 9).

4. **A calibrated failure map** (Section 7): explosion, disjunctive syllogism, and detachment fail with machine-checked finite countermodels; the natural-deduction rules of the source system are derived rules of the calculus, yet the ND system they generate is *provably incomplete* ($\oplus$-self-duality is valid but underivable — Proposition 7.4); $\oplus$ is interderivable with $\wedge$ but not congruent to it, and not definable from the FDE fragment (Proposition 7.5).

5. **A verified generative tier** (Section 8): truth-objects arise from (intensity, structure) channel pairs through an axiomatized, replaceable stability function; independence of the logic from that choice is witnessed by the artifact's import graph, and the limitative facts (unmanifest projection, non-injectivity of initialization) are machine-checked.

Results carry [L: `name`] markers naming the verifying Lean declaration; results not covered by the artifact are marked [paper] and carry full proofs here. The artifact policy is in Appendix A.

2 Syntax

Definition 2.1 (Language)**.** Fix a countably infinite set of atoms $p_0, p_1, \ldots$ (countability is used in Theorem 6.5). Formulas:

$$\varphi \ ::= \ p \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \oplus \varphi.$$

$\varphi \Rightarrow \psi$ abbreviates $\neg\varphi \vee \psi$; it is not a primitive and does not detach (Proposition 7.3). [L: `Nullivance.Syntax.Formula`]

3 Two-channel semantics and threshold signs

Definition 3.1 (Truth-objects; models; valuation)**.** A *truth-object* is a pair $(t, f) \in [0,1]^2$: independent degrees of support for truth and falsity. An *NPL model* is $M = (v, \tau)$ with v assigning a truth-object to every atom and $\tau \in (0,1]$ the *manifestation threshold.* v extends to $V_M(\varphi) = (t_M(\varphi), f_M(\varphi))$ by

$$V_M(\neg\psi) = (f_M(\psi), t_M(\psi)), \qquad V_M(\psi \wedge \chi) = (\min(t,t'), \max(f,f')),$$
$$V_M(\psi \vee \chi) = (\max(t,t'), \min(f,f')), \; V_M(\psi \oplus \chi) = (\min(t,t'), \min(f,f')).$$

writing $(t, f) = V_M(\psi)$, $(t', f') = V_M(\chi)$. The $\{\neg, \wedge, \vee\}$ clauses are the twist-product/bilattice clauses of [5, 8]; $\oplus$ is consensus. [L: `Nullivance.Continuous.TruthObj`, `evalC`]

Definition 3.2 (Signs; states; consequence)**.** $M \models T^+\varphi \iff t_M(\varphi) \geq \tau$; $M \models T^-\varphi \iff t_M(\varphi) < \tau$; $M \models F^+\varphi \iff f_M(\varphi) \geq \tau$; $M \models F^-\varphi \iff f_M(\varphi) < \tau$. The two thresholds classify φ into four *states*: manifest true $\mathbf{T}$, manifest false $\mathbf{F}$, manifest contradiction $\mathbf{B}$ (T^+ and F^+), unmanifest $\mathbf{N}$ (T^- and F^-). Unsigned satisfaction is T^+; designated states are $\{\mathbf{T}, \mathbf{B}\}$. For a set Σ of signed formulas: $\Sigma \models S\varphi$ iff every model (every valuation, every $\tau \in (0,1]$) satisfying Σ satisfies $S\varphi$. [L: `Nullivance.Continuous.SatC`, `Metatheory.ConsequenceC`]

Proposition 3.3 (Threshold invariance)**.** *For every fixed $\tau_0 \in (0,1]$, the relation "every valuation at threshold τ_0 satisfying Σ satisfies $S\varphi$" coincides with $\models$. Quantifying over thresholds in Definition 3.2 is thus a well-definedness statement, not an added strength. [L: `ConsequenceCAt`, `consequenceCAt_iff_consequenceC`]*

Proof sketch. Both lifting directions of Section 6 are uniform in the threshold: a τ_0-model projects to a **4**-valuation satisfying the same signed formulas, and a **4**-valuation embeds at the corners $\{0,1\}^2$, where $\mathbf{1}[1 \geq \tau_0] = 1$ and $\mathbf{1}[0 \geq \tau_0] = 0$ for *every* $\tau_0 \in (0,1]$. Hence each fixed-τ_0 relation equals **4**-consequence (Definition 4.1), hence equals $\models$. □

Lemma 3.4 (Structural facts)**.** *(i) $V_M(\varphi) \in [0,1]^2$ (induction; $[0,1]$ is closed under $\min, \max$). (ii)* Latent collapse*: $V_M(\varphi \oplus \neg\varphi) = (m, m)$ with $m = \min(t_M(\varphi), f_M(\varphi))$ — both channels of the harmonized contradiction agree, so for $\tau > m$ it is in state $\mathbf{N}$. (iii) $\oplus$ is commutative, associative, idempotent, self-dual under $\neg$, with unit $(1,1)$ on the square. (iv) With $(t_1, f_1) \leq_t (t_2, f_2) \iff t_1 \leq t_2 \wedge f_2 \leq f_1$ and $(t_1, f_1) \leq_k (t_2, f_2) \iff t_1 \leq$*

$t_2 \wedge f_1 \leq f_2$: $\wedge/\vee$ are the $\leq_t$ meet/join, $\oplus$ is the $\leq_k$ meet, and $(0,0), (1,1)$ are the $\leq_k$ extrema — the square is a reduct of the product bilattice $[0,1] \odot [0,1]$ [9] with $\oplus$ its consensus; the $\leq_k$-join (gullibility) is deliberately not a connective. [L: `eval_mem`, `latent_collapse`, `oplus2_*`, `le_t/le_k` family]

4 The four-valued matrix and exact projection

Definition 4.1 ($\mathbf{4}$; $\mathbf{4}$-consequence; projection)**.** $\mathbf{4} = \{0,1\}^2 \subseteq [0,1]^2$, corners $\mathbf{T} = (1,0)$, $\mathbf{F} = (0,1)$, $\mathbf{B} = (1,1)$, $\mathbf{N} = (0,0)$, closed under the clauses of Definition 3.1; the $\{\neg, \wedge, \vee\}$ tables are the Belnap–Dunn tables with designated $\{\mathbf{T}, \mathbf{B}\}$ [2, 7], and $\oplus$ restricts to consensus $\otimes$ [1]. A $\mathbf{4}$-valuation w maps atoms to $\mathbf{4}$ and extends by the same clauses; it satisfies $T^+\varphi$ ($F^+\varphi$) iff the first (second) bit of $w(\varphi)$ is 1, and the negative signs are the complements. $\mathbf{4}$-*consequence* $\Sigma \models_{\mathbf{4}} S\varphi$: every $\mathbf{4}$-valuation satisfying Σ satisfies $S\varphi$. The *threshold projection* is $\pi_\tau(x,y) = (\mathbf{1}[x \geq \tau], \mathbf{1}[y \geq \tau])$, where $\mathbf{1}[\cdot]$ is the $\{0,1\}$-valued indicator. [L: `Nullivance.Semantics.V4`, `Metatheory.Consequence4`, `Nullivance.Continuous.proj`]

Theorem 4.2 (Exact projection)**.** *For every model $M = (v, \tau)$ and formula φ: $\pi_\tau(V_M(\varphi)) = V^{\mathbf{4}}_{\pi_\tau \circ v}(\varphi)$. Consequently $M \models S\varphi$ iff the projected $\mathbf{4}$-valuation satisfies $S\varphi$. [L: `exact_projection`, `sat_projection`]*

Proof sketch. Structural induction. The binary cases are the facts $\mathbf{1}[\min(x,y) \geq \tau] = \min(\mathbf{1}[x \geq \tau], \mathbf{1}[y \geq \tau])$ and dually for max — cutworthiness of the standard fuzzy operations [12]; negation commutes with the two cuts because it swaps coordinates. The content is the use: consequence reduces *exactly* to $\mathbf{4}$, with no approximation. □

5 The four-signed analytic calculus

Definition 5.1 (Calculus)**.** A *branch* is a finite set of signed formulas; *closed* iff it contains $S\varphi$ and $\bar{S}\varphi$ for some φ (opposite signs: $\overline{T^+} = T^-$, $\overline{F^+} = F^-$, and conversely $\overline{T^-} = T^+$, $\overline{F^-} = F^+$). The sixteen decomposition rules, one per sign–connective pair (comma = add both to the branch; | = split the branch):

	T^+	T^-	F^+	F^-
$\neg\varphi$	$F^+\varphi$	$F^-\varphi$	$T^+\varphi$	$T^-\varphi$
$\varphi \wedge \psi$	$T^+\varphi, T^+\psi$	$T^-\varphi \mid T^-\psi$	$F^+\varphi \mid F^+\psi$	$F^-\varphi, F^-\psi$
$\varphi \vee \psi$	$T^+\varphi \mid T^+\psi$	$T^-\varphi, T^-\psi$	$F^+\varphi, F^+\psi$	$F^-\varphi \mid F^-\psi$
$\varphi \oplus \psi$	$T^+\varphi, T^+\psi$	$T^-\varphi \mid T^-\psi$	$F^+\varphi, F^+\psi$	$F^-\varphi \mid F^-\psi$

The $\oplus$ row: the T^+ column agrees with $\wedge$ (both truth channels are min), while $F^+(\varphi \oplus \psi)$ is *conjunctive*, unlike the branching $F^+(\varphi \wedge \psi)$ — this is where $\oplus$ and $\wedge$ part ways inside the proof system. $\Sigma \vdash_A S\varphi$ iff some tableau for $\Sigma \cup \{\bar{S}\varphi\}$ closes. The four signs are sets-as-signs [10]: $T^+ = \{\mathbf{T}, \mathbf{B}\}$, $T^- = \{\mathbf{F}, \mathbf{N}\}$, $F^+ = \{\mathbf{F}, \mathbf{B}\}$, $F^- = \{\mathbf{T}, \mathbf{N}\}$. [L: `Nullivance.ProofTheory.Closes`: an inductive closability predicate with two closure and sixteen rule constructors; branching rules take two subderivations]

6 Soundness, completeness, decidability, compactness

Theorem 6.1 (Soundness and completeness over **4**)**.** *For finite Σ: $\Sigma \vdash_A S\varphi \iff \Sigma \models_{\mathbf{4}} S\varphi$. [L: `derives_iff_consequence4`]*

Proof sketch. Soundness: induction on the closability derivation. A closed branch is unsatisfiable since opposite signs are exclusive; each rule preserves satisfiability downward by the local soundness equivalences read off the tables (e.g. the first bit of $w(\varphi \wedge \psi)$ is 1 iff both conjuncts' first bits are). *Completeness*: strong induction on the total size of undecomposed formulas in the branch. Compound members are decomposed by their rule (semantic equivalences justify that unsatisfiability passes to the children); when only literals remain, an unsatisfiable literal branch must contain an opposite-signed pair — else the valuation reading the positive literals off the branch (the canonical valuation) satisfies it. $\square$

Theorem 6.2 (Lifting; continuous completeness)**.** *A finite branch is satisfiable continuously iff satisfiable over **4**. Hence for finite Σ: $\Sigma \vdash_A S\varphi \iff \Sigma \models S\varphi$. [L: `satisfiable_iff_four`, `derives_iff_consequenceC`]*

Proof sketch. Left to right: project the model's valuation through π_τ; satisfaction transfers by Theorem 4.2. Right to left: embed a **4**-valuation at the corners with any threshold (cf. Proposition 3.3); the projection of the embedded model is the original valuation, so satisfaction transfers back. $\square$

Theorem 6.3 (Decidability). *For finite Σ, $\Sigma \vdash_A S\varphi$ (equivalently $\Sigma \models S\varphi$) is decidable. [L: `consequence4Bool_correct`, `decidableDerives`, `decidableConsequenceC`]*

Proof. By Theorems 6.1 and 6.2 it suffices to decide $\models_{\mathbf{4}}$. Evaluation depends only on the atoms occurring in a formula (structural induction — the verified lemma), and the occurring atoms A of $\Sigma \cup \{S\varphi\}$ are finitely many. Hence $\Sigma \models_{\mathbf{4}} S\varphi$ holds iff it holds for the $4^{|A|}$ valuations extending assignments $A \to \mathbf{4}$ by $\mathbf{N}$ elsewhere — a finite computation on the tables. (Alternatively: fair tableau expansion terminates, by the usual subformula/finiteness argument, and decides $\vdash_A$.) □

Proposition 6.4 (Finite complexity). [paper] *Finite signed FOUR consequence is coNP-complete.*

Proof. Non-consequence has a polynomial certificate: assign two bits to each atom occurring in $\Sigma \cup \{S\varphi\}$, evaluate bottom-up, and check that all premises hold while $S\varphi$ fails. For hardness, reduce Boolean tautology. Given a Boolean formula θ, read it as an $\oplus$-free NPL formula θ^* and add, for every variable p_i, the premises $T^+(p_i \vee \neg p_i)$ and $T^-(p_i \wedge \neg p_i)$. These force p_i to be $\mathbf{T}$ or $\mathbf{F}$; $\mathbf{B}$ violates the second premise and $\mathbf{N}$ violates the first. By classical-corner recovery, $\Gamma_\theta \models_{\mathbf{4}} T^+\theta^*$ iff θ is a Boolean tautology. Boolean tautology is coNP-complete by the Cook–Karp theory [6, 11]. □

Theorem 6.5 (Compactness; strong completeness). [L: `compactness_satisfiable4_set`, `compactness_consequence4_set`, `compactness_consequenceC_set`, `derivesSet_iff_consequenceCSet`] *(i) An arbitrary set Σ of signed formulas is $\mathbf{4}$-satisfiable iff every finite subset is. (ii) The same holds for continuous satisfiability. (iii) Reading $\Sigma \vdash_A S\varphi$ for infinite Σ as "some finite subset derives $S\varphi$": $\Sigma \vdash_A S\varphi \iff \Sigma \models S\varphi$ for arbitrary Σ.*

Proof. (i) One direction is restriction. For the converse, fix the enumeration $p_0, p_1, \ldots$ of atoms (Definition 2.1). A *level-n assignment* is $\sigma : \{p_0, \ldots, p_{n-1}\} \to \mathbf{4}$; call σ *good* iff for every finite $\Sigma_0 \subseteq \Sigma$ some $\mathbf{4}$-valuation extending σ satisfies Σ_0. The empty assignment is good (hypothesis). If σ is good, some one-point extension $\sigma \cup \{p_n \mapsto c\}$ is good: otherwise, for each of the four values c pick a finite Σ_c witnessing failure; $\Sigma^* = \bigcup_c \Sigma_c$ is finite, so some $v \supseteq \sigma$ satisfies Σ^*; putting $c := v(p_n)$, v extends $\sigma \cup \{p_n \mapsto c\}$ and satisfies $\Sigma_c \subseteq \Sigma^*$ — contradiction. (The case split over c is exhaustive: $\mathbf{4}$ has exactly four elements.) By recursion along $\mathbb{N}$ choose a chain $\sigma_0 \subseteq \sigma_1 \subseteq \cdots$ of good assignments (countable dependent choice; we work classically). Define $v^*(p_n) := \sigma_{n+1}(p_n)$. For $S\varphi \in \Sigma$: the atoms of φ lie among $p_0, \ldots, p_{n-1}$

for some n; goodness of σ_n with $\Sigma_0 = \{S\varphi\}$ gives $w \supseteq \sigma_n$ satisfying $S\varphi$; w and v^* agree on the atoms of φ, so by finite dependence (Theorem 6.3's lemma) they give φ the same value, and signed satisfaction reads only that value. So v^* satisfies Σ. (ii) Both transfer directions of Theorem 6.2 are memberwise and never use finiteness of the branch: a continuous model's projected valuation satisfies each member, and an embedded **4**-valuation's model satisfies each member. (iii) If some finite $\Sigma_0 \vdash_A S\varphi$ then $\Sigma_0 \models S\varphi$ (Theorem 6.2), hence $\Sigma \models S\varphi$ (shrinking premises weakens the hypothesis of the universally quantified implication). Conversely if no finite subset derives, then by Theorem 6.1 no finite $\Sigma_0 \models_{\mathbf{4}} S\varphi$, so every finite subset of $\Sigma \cup \{\bar{S}\varphi\}$ is satisfiable (failing $S\varphi$ is satisfying $\bar{S}\varphi$); by (i)–(ii) the whole of $\Sigma \cup \{\bar{S}\varphi\}$ is, so $\Sigma \not\models S\varphi$. $\square$

7 The behavioural profile of NPL

Corollary 7.1 (Conservativity; nontriviality)**.** *Unsigned NPL consequence is the all-T^+ fragment of signed consequence (definitional), and on $\oplus$-free formulas it coincides with Belnap–Dunn FDE consequence: identical value space, operations, and designated set define the same matrix consequence* [paper]. *Nontriviality: $\emptyset \nvdash_A T^+p$, by the constant-$\mathbf{N}$ countermodel [L:* `consistency_witness`*]; note every formula evaluates to $\mathbf{N}$ under the constant-$\mathbf{N}$ valuation, so NPL has no theorems — like FDE.*

Corollary 7.2 (Paraconsistency)**.** *$\{T^+p,\ T^+\neg p\} \not\models T^+q$: at $p \mapsto \mathbf{B}$, $q \mapsto \mathbf{N}$ both premises hold ($t(p) = f(p) = 1$) and the conclusion fails ($t(q) = 0$). Explosion, ex falso, and disjunctive syllogism fail. [L:* `non_explosion`*]*

Proposition 7.3 (The material conditional does not detach)**.** *$\{T^+p,\ T^+(p \Rightarrow q)\} \not\models T^+q$: same witness, $t(\neg p \vee q) = \max(f(p), t(q)) = \max(1, 0) = 1$ yet $t(q) = 0$. Consequently $\vee$-elimination must be taken as a meta-rule (reasoning by cases), not routed through $\Rightarrow$. [L:* `modus_ponens_fails`, `disj_elim`*]*

Proposition 7.4 (Derived rules; ND incompleteness)**.** *Let $\vdash_{\mathrm{ND}}$ be the calculus generated by exactly the following rules (finite premise lists; the source system's rule set): assumption; $\wedge$-introduction and both eliminations; both $\vee$-introductions; $\vee$-elimination by cases (from $\Gamma \vdash \varphi \vee \psi$, $\varphi, \Gamma \vdash \chi$, $\psi, \Gamma \vdash \chi$ infer $\Gamma \vdash \chi$); $\neg\neg$-introduction and elimination; the four De Morgan directions for $\wedge$ and $\vee$; $\oplus$-introduction and both $\oplus$-eliminations. Then: (a) every $\vdash_{\mathrm{ND}}$-rule is a derived rule of $\vdash_A$ (each is semantically valid — the $\oplus$-rules because the truth channel of $\oplus$ is* min *— and Theorem 6.1 packages validity* *as derivability); but (b)* $\vdash_{\mathrm{ND}}$ *is* incomplete*:* $\oplus$*-self-duality* $\neg(p\oplus q) \models \neg p\oplus\neg q$ *(a value identity, Lemma 3.4(iii)) is not* $\vdash_{\mathrm{ND}}$*-derivable. For (b): no rule of* $\vdash_{\mathrm{ND}}$ *constrains the falsity channel of* $\oplus$ *— on the truth channel* $\oplus$ *and* $\wedge$ *agree — so reading* $\oplus$ *as* $\wedge$ *throughout preserves the soundness of every rule; but the target's* $\wedge$*-reading,* $\neg(p \wedge q) \models \neg p \wedge \neg q$*, fails at* $p \mapsto \mathbf{T}$*,* $q \mapsto \mathbf{F}$*. Self-duality is a falsity-channel fact, and the four-signed calculus is canonical precisely because its* $F^{\pm}\oplus$ *rules reach that channel. [L:* `ProofTheory.ND` *(16 constructors),* `ND.sound_w` *(generic truth-channel soundness),* `nd_sound`*,* `nd_incomplete`*]*

Proposition 7.5 ($\oplus$ is not definable from $\{\neg, \wedge, \vee\}$)**.** *The* $\{\neg, \wedge, \vee\}$ *fragment is classically closed: a* $\oplus$*-free formula takes a classical value (*$\mathbf{T}/\mathbf{F}$*) whenever its atoms do (structural induction on the corner tables). Since* $\mathbf{T}\oplus\mathbf{F} = \mathbf{N}$ *is not classical, no* $\oplus$*-free formula defines* $\oplus$*: harmonization strictly extends the expressive power of the FDE fragment. [L:* `classical_closed`*,* `oplus_not_definable`*]*

Proposition 7.6 (Exact role of $\oplus$; non-congruence)**.** $T^{+}(\varphi \oplus \psi)$ *and* $T^{+}(\varphi \wedge \psi)$ *coincide in every model (both truth channels are* min*), so the two formulas are interderivable; yet at* $\varphi \mapsto \mathbf{B}, \psi \mapsto \mathbf{T}$*:* $\neg(\varphi \wedge \psi)$ *is designated (*$\neg\mathbf{B} = \mathbf{B}$*) while* $\neg(\varphi\oplus\psi)$ *is not (*$\neg\mathbf{T} = \mathbf{F}$*). Interderivability does not license replacement under* $\neg$*; only value equivalence does. NPL is non-congruential by design. [L:* `conj_oplus_interderivable`*,* `oplus_conj_Fpos_fails`*,* `noncongruence_witness`*]*

Corollary 7.7 (Latent collapse, thresholded)**.** $V_M(\varphi\oplus\neg\varphi) = (m, m)$ *with* $m = \min(t, f)$ *(Lemma 3.4(ii)); for* $\tau > m$ *the harmonized contradiction is unmanifest (*$\mathbf{N}$*), not explosive. [L:* `latent_collapse`*]*

8 The generative tier

Nothing in Sections 2 to 7 depends on how atoms acquire truth-objects; in the artifact this is a checkable property of the import graph (the module `Generative` is imported by none of the core modules). This section adds the initialization theory.

Definition 8.1 (Generative frame; channels; quasivance)**.** A *generative frame* is $F = (d, \Phi)$ with $d \geq 1$ and $\Phi : [0, 1]^d \to [0, 1]$ satisfying $\Phi(1-\Theta) = \Phi(\Theta)$ and $\Phi(\frac{1}{2}, \ldots, \frac{1}{2}) = 1$. A *support channel* $c = (\alpha, \Theta)$ has *existence intensity* $\alpha \in [0, 1]$ and *polarization structure* $\Theta \in [0, 1]^d$; its effective intensity is $\alpha \cdot \Phi(\Theta)$. Two independent channels per atom (for/against) induce the truth-object $(\alpha_T\Phi(\Theta_T),\ \alpha_F\Phi(\Theta_F)) \in [0,1]^2$ (interface theorem: products of $[0,1]$-values stay in $[0,1]$), so the core applies to any admissible Φ; the canonical instance $\Phi_c(\Theta) = (\prod_k(1-2|\Theta_k - \frac{1}{2}|))^{1/d}$ is verified admissible. A channel is *quasivant* iff $\alpha = 0$ with non-neutral Θ: unmanifest yet structured — the notion that motivated the system (structure at probability zero is invisible to entropy-based measures [17]). [L: `GenFrame`, `GenState.init_mem`, `canonFrame`, `Channel.Quasivant`]

Proposition 8.2 (Quasivance at the interface)**.** *(i) A quasivant state initializes to* $(0,0)$ *and projects to* $\mathbf{N}$ *for every* $\tau \in (0,1]$*. (ii)* Forgetfulness*: initialization is not injective; quasivant and non-quasivant states can initialize identically, so quasivance is not recoverable from the truth-object — claims about* $\mathbf{N}$ *in the core concern unmanifestness only. (iii) In the canonical frame, one fully polarized component annihilates effective intensity even at* $\alpha = 1$*. (iv) Initialization is flip-covariant and surjective onto the square. We state (ii) prominently: it is the honest price of the generative reading. [L: `quasivant_projects_N`, `init_not_injective`, `polar_kills_intensity`, `init_flip`, `init_surjective`]*

9 Related work

Belnap–Dunn / FDE. The $\mathbf{4}$ fragment on $\{\neg,\wedge,\vee\}$ *is* FDE [2, 3, 7]; see [13] for the landscape of FDE extensions, within which NPL sits as a consensus expansion with a projection layer. **Logical bilattices.** The square is the product bilattice [9, 8]; $\oplus$ is consensus; complete Gentzen/Hilbert calculi for the four-valued bilattice language are due to Arieli–Avron [1] (see also [16]). The precise relation:

Proposition 9.1 (Position within the bilattice framework)**.** [paper] *(i) For every* $\tau \in (0,1]$*,* $D_\tau = \{(t,f) : t \geq \tau\}$ *is a prime bifilter of* $[0,1] \odot [0,1]$ *(the five defining conditions hold componentwise on the truth channel), so* $\langle [0,1] \odot [0,1], D_\tau \rangle$ *is a logical bilattice; by the Arieli–Avron collapse theorem (every logical bilattice determines the same consequence as* $\langle \mathbf{4}, \{t, \top\} \rangle$ *— [1, Thm. 2.17], as presented in [16, Thm. 2.1.4]), the* unsigned *NPL consequence is exactly the* $\{\wedge,\vee,\otimes,\neg\}$*-fragment of their logic* LB *— subsumed, and consistent with our independent Theorem 6.2. (ii) The satisfaction set of* T^-*,* $\{(t,f) : t < \tau\}$*, contains* $(0,0)$ *but not* $(1,1)$*, hence is not upward closed in* $\leq_k$ *and is not a bifilter (bifilters are upward closed in both orders and contain the knowledge top). The four-signed relation is therefore not a logical-bilattice matrix consequence; the signed layer is where NPL leaves the* *framework. (iii) Nor can the negative signs be internalized: no formula ψ satisfies $T^+\psi \iff T^-p$ in all valuations, since the constant-$\mathbf{B}$ valuation is a fixpoint of every connective — every formula is designated there, while T^-p fails. [L:* `signs_not_internalizable`*, via* `eval_const_B`*]*

Logic programming. Fitting [8] interprets logic programs over continuous bilattices with the knowledge operations (including consensus) available in rule bodies; that line provides fixed-point semantics for programs, not a consequence relation with a completeness theorem for a propositional language — the two uses of the same algebra are complementary. **Paraconsistent Gödel logic.** The continuous $\{\neg, \wedge, \vee\}$ clauses coincide with the propositional base of [5] (twist product of $[0,1]$); their language has a Gödel implication and no consensus, ours has consensus and no implication, and the consequence relations differ (signed vs. order-based). **Fuzzy sets.** The projection core is α-cut cutworthiness [12]. **Tableaux.** Sets-as-signs tableaux are standard [10]; FDE has textbook signed tableaux [15]. The closest tableau precedent for our setting is the *constraint tableaux* of [4] for two-dimensional Łukasiewicz and Gödel logics over the same twist product of $[0,1]$: there, branches carry order constraints on real values because the fuzzy implications do not reduce to any finite matrix. Our calculus needs only four *finite* signs precisely because NPL has no implication — all connectives are cut-worthy, exact projection holds, and consequence collapses onto **4** (Theorem 4.2); adding a Gödel implication to NPL would break that theorem. The two designs are complementary solutions for different languages. **LP.** Designating **B** descends from [14]. **Verified metatheory.** Paraconsistent logics have been formalized in proof assistants before — notably an infinite-valued paraconsistent logic with its metatheory in Isabelle [18]; we are not aware of verified *completeness* for a consensus-bearing logic over $[0,1]^2$.

To our knowledge — search scope documented in the project's related-work audit; last sweep July 2026, covering post-2022 twist-product, bilattice-calculus, and proof-assistant literature — no prior system combines a consensus connective over $[0,1]^2$ with four-signed consequence and a machine-checked completeness theorem.

10 Conclusion and open problems

NPL's propositional development is closed end to end: generative initialization, continuous two-channel semantics, exact projection, a four-signed calculus, soundness/completeness/decidability verified in Lean, compactness, strong completeness, classical-corner recovery verified in Lean, coNP-complete finite consequence, a provably incomplete ND fragment explaining why the signed calculus is canonical, and a precise statement of which parts of the system are and are not instances of the logical-bilattice framework. Since this manuscript snapshot, the project has also Lean-verified completeness for the $\oplus$-De-Morgan extension of $\vdash_{\mathrm{ND}}$ in the repository docs. Open: (1) the finite-domain quantified proof theory and its completeness theorem; (2) the full infinite-domain quantified extension; the emergence layer of the source material remains outside the formal development by design.

A The Lean artifact

Availability. The `Nullivance` library will be archived with a DOI at submission; until then it is available from the author on request. **Toolchain.** Lean 4 (`leanprover/lean4:v4.32.0-rc1`) with mathlib pinned via the Lake manifest; build with `lake exe cache get && lake build` (approx. 2000 jobs; the library itself compiles in under a minute on commodity hardware). **Coverage policy.** The build is `sorry`-free. Every result marked [L] is verified as stated, with the following systematic correspondences: branches are lists (finite sets up to the verified monotonicity lemma); "some tableau closes" is the inductive predicate `Closes` (two closure, sixteen rule constructors); the calculus-level results are stated against `Closes` directly; explicit finite tableau proof trees are equivalent to `Closes`. Results marked [paper] — Corollary 7.1 (FDE-conservativity half), Proposition 6.4, Proposition 9.1 — carry full proofs in this paper and are not formalized; the termination and fair-search notions appear only in the alternative proof remark of Theorem 6.3. **Modules.** `Syntax`, `Semantics` (**4**), `Continuous` (square, projection, bilattice orders), `ProofTheory` (`Closes`, `ND`, `NDO`), `Tableau` (explicit proof trees), `Metatheory` (soundness, completeness, lifting, τ-invariance, behavioural profile), `Decidability` (finite model checker and `Decidable` instances), `Compactness` (Set/Finset API, compactness, strong completeness), `Classical` (Boolean recovery on the glut/gap-free fragment), `FiniteFO` (finite-domain quantified first pass), `Generative` (Section 8; imported by no core module).

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